

Literature Review

LPS / endotoxin

Measurement of endotoxin activity in critically ill patients using whole blood neutrophil dependent chemiluminescence

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EAA outperforms LAL for clinical endotoxin detection

The Endotoxin Activity Assay (EAA) enables rapid (<30 min), whole-blood detection of endotoxemia using <100 µL sample volume and shows high specificity for LPS (validated by polymyxin B inhibition), overcoming major limitations of the conventional LAL assay caused by circulating inhibitors and non-specific fungal interference.

Endotoxemia is a reliable diagnostic marker of Gram-negative sepsis

All patients with culture-proven Gram-negative infections exhibited endotoxin levels >50 pg/mL (100% sensitivity in this cohort), with significantly higher levels in sepsis compared to non-sepsis patients; endotoxemia correlated with leukocytosis but not mortality, indicating strong diagnostic value rather than prognostic significance.

Relationship between Plasma Levels of Lipopolysaccharide (LPS) and LPS-Binding Protein in Patients with Severe Sepsis and Septic Shock

Steven M. Opal , Patrick J. Scannan, Jean-Louis Vincent, Mark White, Stephen F. Carroll, John E. Palardy, Nicolas A. Parejo, John P. Pribble, Jon H. Lemke

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Endotoxin (LPS) levels correlate with sepsis severity and mortality

Healthy: <10 pg/mL → Severe sepsis: 110–726 pg/mL (median ~300 pg/mL) → Non-survivors: ~515 pg/mL; endotoxemia present in ~78% of severe sepsis cases, with levels >50 pg/mL indicating high-risk infection and increased mortality, especially in elderly and septic shock patients.

LBP reflects host inflammatory capacity, not endotoxin burden

Healthy: ~2–6 µg/mL → Sepsis: ~10–60 µg/mL (median ~31 µg/mL); survivors show higher LBP (~33 µg/mL) than non-survivors (~28 µg/mL), indicating preserved hepatic acute-phase response; LBP and endotoxin levels are uncorrelated, representing independent biological axes.

Elevated Levels of Lipopolysaccharide-Binding Protein and Soluble CD14 in Plasma in Neonatal Early-Onset Sepsis

Authors: Reinhard Berner , Birgitt Füll, Felix Stelter, Jana Dröse, Hans-Peter Müller, Christine Schütt | [AUTHORS INFO & AFFILIATIONS](#)

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LBP and sCD14 are significantly elevated at birth in neonatal sepsis and persist beyond 24 h

Septic neonates showed markedly higher plasma **LBP** (median **36.6 µg/mL**; IQR ~**10.6–50.4 µg/mL**) compared to healthy controls (median **7.8 µg/mL**; IQR ~**6.1–10.6 µg/mL**; **P < 0.001**), with levels further increasing at 24–48 h (median ~**60 µg/mL**); **sCD14** was also elevated (median ~**0.42 µg/mL** vs **0.28 µg/mL** in controls; **P < 0.001**) but remained stable over time.

LBP and sCD14 reflect innate immune activation independent of pathogen type

Elevated **LBP** (~**20–60 µg/mL**) and **sCD14** (~**0.35–0.65 µg/mL**) were observed regardless of blood culture positivity or Gram status (including *Streptococcus agalactiae*), and strongly correlated with pro-inflammatory cytokines (IL-6, IL-8, IL-1β, G-CSF), indicating activation of the neonatal innate immune response rather than Gram-negative-specific endotoxemia.

Perioperative change of plasma endotoxin levels in early infants

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Endotoxin levels in immunocompromised children with fever

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Defined upper limit of normal endotoxin in children

Plasma endotoxin levels in healthy and convalescent children were **≤35 pg/mL (0.10 EU/mL)**, establishing **35 pg/mL as the pediatric upper normal threshold** for endotoxemia.

Endotoxemia contributes to unexplained fever without bacteremia

14% (5/36) of immunocompromised children with febrile episodes showed **elevated endotoxin levels >35 pg/mL** despite absence of bacteremia or clinically diagnosed infection; endotoxin levels normalized during convalescence, implicating transient endotoxemia as a **potential pyrogenic mechanism** in unexplained pediatric fever.

Endotoxin-specific chromogenic assay for plasma in pregnant women, umbilical cords, neonates and children

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Physiological endotoxin levels are very low in normal pregnancy, neonates, and children

Using an endotoxin-specific chromogenic assay (Endospecky), endotoxin levels were **0.8 ± 0.4 pg/mL** in maternal blood and **1.3 ± 0.6 pg/mL** in umbilical cord blood after normal delivery; non-infectious neonates showed **$\sim 6.7 \pm 5.1$ pg/mL**, and healthy children had levels **<10 pg/mL**, defining a low baseline endotoxin range in pediatric populations.

Direct detection of bacteremia by exploiting host-pathogen interactions of lipoteichoic acid and lipopolysaccharide

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LPS and LTA are validated, pathogen-proximal biomarkers ideally suited for SERS detection

Direct detection of **lipopolysaccharide (LPS)** and **lipoteichoic acid (LTA)** captures conserved bacterial **PAMPs** that circulate in blood during bacteremia even at **very low bacterial loads (~1 CFU/mL)**; this supports **SERS-based, label-free detection** of molecular fingerprints rather than indirect host response markers or activity-dependent endotoxin assays.

Quantification of Lipoteichoic Acid in Hemodialysis Patients With Central Venous Catheters

[Amy Barton Pai](#)^{1,*}, [Adinoyi Garba](#)², [Paul Neumann](#)³, [Alexander J Prokopienko](#)⁴, [Gabrielle Costello](#)¹, [Michael C Dean](#)¹, [Sriram Narsipur](#)⁵

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LTA is quantifiable systemically and reflects Gram-positive biofilm burden beyond culture detection

Lipoteichoic acid (LTA), a Gram-positive cell wall component, was detected in both catheter aspirate and plasma of hemodialysis patients with central venous catheters, with **plasma LTA 3.93 ng/mL (0.25–15)** exceeding **catheter aspirate LTA 2.38 ng/mL (0.1–8.1)** in **70% of patients (p = 0.01)**, demonstrating continuous biofilm shedding and systemic dissemination not captured by routine culture.

Functionalized arrays of Raman-enhancing nanoparticles for capture and culture-free analysis of bacteria in human blood

Ting-Yu Liu ¹, Kun-Tong Tsai, Huai-Hsien Wang, Yu Chen, Yu-Hsuan Chen, Yuan-Chun Chao, Hsuan-Hao Chang, Chi-Hung Lin, Juen-Kai Wang, Yuh-Lin Wang

Affiliations + expand

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Vancomycin-functionalized SERS substrates enable culture-free capture and direct Raman detection of whole bacteria from blood

Vancomycin-coated Ag nanoparticle arrays exploit specific hydrogen bonding with bacterial peptidoglycan, achieving up to **~1,000-fold enhancement in bacterial capture** while **excluding blood cells**, enabling **label-free, rapid SERS analysis directly from complex matrices like human blood**.

Chemical functionalization enhances SERS sensitivity and enables phenotypic discrimination

Vancomycin coating not only increases bacterial adhesion but also pulls bacterial cell walls into SERS “hot spots,” resulting in **400–500% higher Raman signal** and allowing **spectral differentiation between vancomycin-susceptible and resistant strains**, demonstrating SERS as a powerful platform for **rapid diagnostics and antimicrobial susceptibility assessment without culture**.

The biochemical origins of the surface enhanced Raman spectra of bacteria: metabolomics profiling by SERS

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Bacterial SERS signals originate from extracellular purine metabolites, not cell-wall components

At 785 nm excitation, SERS spectra of whole bacteria are dominated by **purine degradation products**—adenine, hypoxanthine, xanthine, guanine, uric acid, and AMP—released during **starvation induced by water washing**. Identical SERS fingerprints from whole cells and cell-free supernatants confirm that **peptidoglycan (NAG/NAM), lipids, proteins, or LPS do not contribute significantly** to these spectra.

785 nm SERS functions as a metabolic stress readout rather than a structural biomarker

The bacterial SERS fingerprint reflects **species-specific purine metabolism and enzyme availability**, not Gram type or cell-envelope architecture. Signal intensity and composition are strongly influenced by **sample preparation, washing buffer, and starvation duration**, establishing SERS as a **metabolic biomarker platform sensitive to bacterial stress state**, rather than a direct detector of surface antigens or cell-wall molecules.

Escherichia coli research on Raman measurement mechanism and diagnostic model

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Raman/SERS signal origin is mechanistically linked to bacterial cell-wall chemistry

Raman/SERS signatures of *E. coli* can be **physically and biologically anchored to peptidoglycan biomarkers**, as Density Functional Theory (DFT)–validated vibrational modes of **N-acetylmuramic acid (MurNAc)** and **N-acetylglucosamine (GlcNAc)** directly explain the origin of diagnostic Raman bands, enabling **mechanism-based interpretation rather than empirical fingerprinting**.

Key Finding 2 — Mechanism-informed feature selection significantly enhances diagnostic accuracy

Incorporating **DFT-certified MurNAc/GlcNAc Raman peaks (3010–3029 cm⁻¹)** with chemometric selection (SPA) and machine learning (Random Forest) **substantially reduces misdiagnosis** and achieves **near-clinical performance (~98% accuracy, AUC ≈ 1)**, demonstrating that **biomarker-guided Raman/SERS models outperform untargeted spectral classification**.

Comparison table for Which Solvent Has the *Highest Overall Solubility*?

Biomarker / Solvent	Water	PBS (pH 7.2–7.4)	DMSO	DMF	Ethanol
LTA (Staph. aureus)	5 mg/mL (hazy) (MilliporeSigma)	Moderate (hazy; comparable to water), <i>qualitative</i> (MilliporeSigma)	Unknown	Unknown	Moderate (MilliporeSigma)
MurNAc	~50 mg/mL (clear) (ChemicalBook)	~10 mg/mL (Cayman Chemical)	~30 mg/mL (Cayman Chemical)	~2.5 mg/mL (Cayman Chemical)	Unknown
LPS (E. coli O111:B4)	~5 mg/mL (hazy; up to ~20 mg/mL with heat) (MilliporeSigma)	Moderate (hazy solutions in PBS; similar to water) (MilliporeSigma)	Poor / Not reported (organic solvents do not yield clearer solutions) (MilliporeSigma)	Unknown	Poor (methanol/ethanol produce turbid suspensions with precipitates) (MilliporeSigma)

Best Solvent for Each Biomarker

Biomarker	Best Solvent /Highest Solubility
LTA	Water (5 mg/mL)
MurNAc	Water (~50 mg/mL)
LPS	Water (~5 mg/mL)